In [1]:
```
#### Import python libraries
import psycopg2
```

In [2]:
```
import pandas as pd
```

In [3]:
```
df = pd.read_csv(r"C:\Users\hp\Desktop\Datasets\sales.csv")
```

## Extract

In [4]:
```
def Extract():
    df = pd.read_csv(r"C:\Users\hp\Desktop\Datasets\sales.csv")
    return df
```

## Transform

In [5]:
```
df.duplicated().sum()
```

Out[5]:
```
np.int64(0)
```

In [6]:
```
def Transform():
    df.drop('ADDRESSLINE2',axis = 1, inplace = True)
    return df
```

In [7]:
```
df[df['STATE'].isnull()].head()
```

Out[7]:

| | ORDERNUMBER | QUANTITYORDERED | PRICEEACH | ORDERLINENUMBER | SALES | ORDER |
|---|---|---|---|---|---|---|
| **1** | 10121 | 34 | 81.35 | 5 | 2765.90 | 5/7 |
| **2** | 10134 | 41 | 94.74 | 2 | 3884.34 | 7/1 |
| **6** | 10180 | 29 | 86.13 | 9 | 2497.77 | 11/11 |
| **7** | 10188 | 48 | 100.00 | 1 | 5512.32 | 11/18 |
| **9** | 10211 | 41 | 100.00 | 14 | 4708.44 | 1/15 |

5 rows × 25 columns

In [8]:
```
df[df['STATE'].notnull()].head()
```

Out[8]:

| | ORDERNUMBER | QUANTITYORDERED | PRICEEACH | ORDERLINENUMBER | SALES | ORDER |
|---|---|---|---|---|---|---|
| **0** | 10107 | 30 | 95.70 | 2 | 2871.00 | 2/24 |
| **3** | 10145 | 45 | 83.26 | 6 | 3746.70 | 8/25 |
| **4** | 10159 | 49 | 100.00 | 14 | 5205.27 | 10/10 |
| **5** | 10168 | 36 | 96.66 | 1 | 3479.76 | 10/28 |
| **8** | 10201 | 22 | 98.57 | 2 | 2168.54 | 12/1 |

5 rows × 25 columns

In [9]:
```
df.groupby('COUNTRY')['STATE'].apply(lambda x: x.isnull().mean())
```

Out[9]:
```
COUNTRY
Australia      0.000000
Austria        1.000000
Belgium        1.000000
Canada         0.000000
Denmark        1.000000
Finland        1.000000
France         1.000000
Germany        1.000000
Ireland        1.000000
Italy          1.000000
Japan          0.000000
Norway         1.000000
Philippines    1.000000
Singapore      1.000000
Spain          1.000000
Sweden         1.000000
Switzerland    1.000000
UK             0.819444
USA            0.000000
Name: STATE, dtype: float64
```

In [10]:
```
df['STATE'] = df['STATE'].fillna('Unknown')
```

In [11]:
```
print(df.isnull().sum())
```

```
ORDERNUMBER           0
QUANTITYORDERED       0
PRICEEACH             0
ORDERLINENUMBER       0
SALES                 0
ORDERDATE             0
STATUS                0
QTR_ID                0
MONTH_ID              0
YEAR_ID               0
PRODUCTLINE           0
MSRP                  0
PRODUCTCODE           0
CUSTOMERNAME          0
PHONE                 0
ADDRESSLINE1          0
ADDRESSLINE2       2521
CITY                  0
STATE                 0
POSTALCODE           76
COUNTRY               0
TERRITORY          1074
CONTACTLASTNAME       0
CONTACTFIRSTNAME      0
DEALSIZE              0
dtype: int64
```

In [12]:
```
pd.crosstab(df['COUNTRY'], df['TERRITORY'])
```

Out[12]:

| TERRITORY<br>COUNTRY | APAC | EMEA | Japan |
|---|---|---|---|
| **Australia** | 185 | 0 | 0 |
| **Austria** | 0 | 55 | 0 |
| **Belgium** | 0 | 33 | 0 |
| **Denmark** | 0 | 63 | 0 |
| **Finland** | 0 | 92 | 0 |
| **France** | 0 | 314 | 0 |
| **Germany** | 0 | 62 | 0 |
| **Ireland** | 0 | 16 | 0 |
| **Italy** | 0 | 113 | 0 |
| **Japan** | 0 | 0 | 52 |
| **Norway** | 0 | 85 | 0 |
| **Philippines** | 0 | 0 | 26 |
| **Singapore** | 36 | 0 | 43 |
| **Spain** | 0 | 342 | 0 |
| **Sweden** | 0 | 57 | 0 |
| **Switzerland** | 0 | 31 | 0 |
| **UK** | 0 | 144 | 0 |

In [13]:
```
df[df['COUNTRY'] == 'Singapore']
```

Out[13]:

| | ORDERNUMBER | QUANTITYORDERED | PRICEEACH | ORDERLINENUMBER | SALES | O |
|---|---|---|---|---|---|---|
| **30** | 10150 | 45 | 100.00 | 8 | 10993.50 | |
| **137** | 10150 | 20 | 100.00 | 1 | 3191.20 | |
| **143** | 10217 | 48 | 100.00 | 4 | 7020.48 | |
| **146** | 10259 | 26 | 100.00 | 12 | 4033.38 | |
| **187** | 10117 | 33 | 100.00 | 9 | 6034.38 | |
| **...** | ... | ... | ... | ... | ... | |
| **2510** | 10117 | 38 | 79.68 | 6 | 3027.84 | |
| **2618** | 10150 | 20 | 100.00 | 3 | 2104.00 | |
| **2624** | 10217 | 31 | 88.00 | 6 | 2728.00 | |
| **2718** | 10117 | 45 | 83.42 | 1 | 3753.90 | |
| **2798** | 10117 | 50 | 43.68 | 2 | 2184.00 | |

79 rows × 25 columns

In [14]:
```
df[df['TERRITORY'].isnull()] ['COUNTRY'].value_counts()
```

Out[14]:
```
COUNTRY
USA       1004
Canada      70
Name: count, dtype: int64
```

In [15]:
```
pd.crosstab(df['COUNTRY'], df['TERRITORY'], dropna=False)
```

Out[15]:

| TERRITORY | APAC | EMEA | Japan | NaN |
|---|---|---|---|---|
| COUNTRY | | | | |
| Australia | 185 | 0 | 0 | 0 |
| Austria | 0 | 55 | 0 | 0 |
| Belgium | 0 | 33 | 0 | 0 |
| Canada | 0 | 0 | 0 | 70 |
| Denmark | 0 | 63 | 0 | 0 |
| Finland | 0 | 92 | 0 | 0 |
| France | 0 | 314 | 0 | 0 |
| Germany | 0 | 62 | 0 | 0 |
| Ireland | 0 | 16 | 0 | 0 |
| Italy | 0 | 113 | 0 | 0 |
| Japan | 0 | 0 | 52 | 0 |
| Norway | 0 | 85 | 0 | 0 |
| Philippines | 0 | 0 | 26 | 0 |
| Singapore | 36 | 0 | 43 | 0 |
| Spain | 0 | 342 | 0 | 0 |
| Sweden | 0 | 57 | 0 | 0 |
| Switzerland | 0 | 31 | 0 | 0 |
| UK | 0 | 144 | 0 | 0 |
| USA | 0 | 0 | 0 | 1004 |

In [16]:
```
df.loc[
    (df['COUNTRY'].isin(['USA', 'Canada'])) &
    (df['TERRITORY'].isna()), 'TERRITORY'] = 'North America'
```

In [17]:
```
df[df['COUNTRY'].isin(['USA', 'Canada'])] [['COUNTRY', 'TERRITORY']].head(20)
```

Out[17]:

| | COUNTRY | TERRITORY |
|---|---|---|
| **0** | USA | North America |
| **3** | USA | North America |
| **4** | USA | North America |
| **5** | USA | North America |
| **8** | USA | North America |
| **11** | USA | North America |
| **12** | USA | North America |
| **13** | USA | North America |
| **15** | USA | North America |
| **18** | USA | North America |
| **19** | USA | North America |
| **23** | USA | North America |
| **29** | USA | North America |
| **31** | USA | North America |
| **33** | USA | North America |
| **35** | Canada | North America |
| **36** | USA | North America |
| **37** | USA | North America |
| **38** | USA | North America |
| **44** | USA | North America |

In [18]:
```
df.loc[
    df['COUNTRY'].isin(['USA', 'Canada']), 'TERRITORY'].isna().sum()
```

Out[18]:
```
np.int64(0)
```

In [19]:
```
df[df['TERRITORY'].isna()] ['COUNTRY'].value_counts()
```

Out[19]:
```
Series([], Name: count, dtype: int64)
```

In [20]:
```
df['TERRITORY'].isna().sum()
```

Out[20]:
```
np.int64(0)
```

In [21]:
```
df[df['COUNTRY'].isin(['USA', 'Canada'])] ['TERRITORY'].value_counts(dropna=False)
```

Out[21]:
```
TERRITORY
North America    1074
Name: count, dtype: int64
```

In [22]:
```
df[df['TERRITORY'].isna()] ['COUNTRY'].value_counts()
```

Out[22]:
```
Series([], Name: count, dtype: int64)
```

In [23]:
```
df['ORDERDATE'] = pd.to_datetime(df['ORDERDATE'])
```

In [24]:
```
df['Total_Revenue'] = df['QUANTITYORDERED'] * df['PRICEEACH']
```

In [25]:
```
df = df[df['YEAR_ID'] == 2004]
```

In [26]:
```
df.rename(columns={'Total_Revenue': 'TOTAL_REVENUE'}, inplace=True)
```

In [27]:
```
df.columns
```

Out[27]:
```
Index(['ORDERNUMBER', 'QUANTITYORDERED', 'PRICEEACH', 'ORDERLINENUMBER',
       'SALES', 'ORDERDATE', 'STATUS', 'QTR_ID', 'MONTH_ID', 'YEAR_ID',
       'PRODUCTLINE', 'MSRP', 'PRODUCTCODE', 'CUSTOMERNAME', 'PHONE',
       'ADDRESSLINE1', 'ADDRESSLINE2', 'CITY', 'STATE', 'POSTALCODE',
       'COUNTRY', 'TERRITORY', 'CONTACTLASTNAME', 'CONTACTFIRSTNAME',
       'DEALSIZE', 'TOTAL_REVENUE'],
      dtype='object')
```

In [28]:
```
pd.set_option('display.max_columns', None)
df.head()
```

Out[28]:

| | ORDERNUMBER | QUANTITYORDERED | PRICEEACH | ORDERLINENUMBER | SALES | ORDE |
|---|---|---|---|---|---|---|
| **9** | 10211 | 41 | 100.0 | 14 | 4708.44 | 200 |
| **10** | 10223 | 37 | 100.0 | 1 | 3965.66 | 200 |
| **11** | 10237 | 23 | 100.0 | 7 | 2333.12 | 200 |
| **12** | 10251 | 28 | 100.0 | 2 | 3188.64 | 200 |
| **13** | 10263 | 34 | 100.0 | 2 | 3676.76 | 200 |

## LOAD

In [34]:
```
fil_path = "sales.csv"

def Extract():
    df = pd.read_csv(r"C:\Users\hp\Desktop\Datasets\sales.csv")
```

```
    return df

def Transform(df):
    df.drop('ADDRESSLINE2',axis = 1, inplace = True)
    df['STATE'] = df['STATE'].fillna('Unknown')
    df.loc[
        (df['COUNTRY'].isin(['USA', 'Canada'])) &
        (df['TERRITORY'].isna()), 'TERRITORY'] = 'North America'
    df['ORDERDATE'] = pd.to_datetime(df['ORDERDATE'])
    df['Total_Revenue'] = df['QUANTITYORDERED'] * df['PRICEEACH']
    df = df[df['YEAR_ID'] == 2004]
    df.rename(columns={'Total_Revenue': 'TOTAL_REVENUE'}, inplace=True)
    return df


def load(df):
    connection = psycopg2.connect(
    host = "localhost",
    dbname = "Sales ",
    user = "postgres",
    password = "1234"

    )

    con = connection.cursor()

    for _, row in df.iterrows():
        con.execute("""
            INSERT INTO Sales_Record(
            ORDERNUMBER, QUANTITYORDERED, PRICEEACH, ORDERLINENUMBER,
           SALES, ORDERDATE, STATUS, QTR_ID, MONTH_ID, YEAR_ID,
           PRODUCTLINE, MSRP, PRODUCTCODE, CUSTOMERNAME, PHONE,
           ADDRESSLINE1, CITY, STATE, POSTALCODE, COUNTRY, TERRITORY,
           CONTACTLASTNAME, CONTACTFIRSTNAME, DEALSIZE, TOTAL_REVENUE)
            VALUES (%s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s,
            ON CONFLICT (ORDERNUMBER) DO NOTHING """, tuple (row)
           )

    connection.commit()
    con.close()
    connection.close()

if __name__ == "__main__":
    try:
        raw_data = Extract()
        clean_data = Transform(raw_data)
        load(clean_data)
    except Exception as e:
        print(f"Error: {e}")
```

```
C:\Users\hp\AppData\Local\Temp\ipykernel_14628\1651861553.py:16: SettingWithCopyWarn
ing:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/u
ser_guide/indexing.html#returning-a-view-versus-a-copy
  df.rename(columns={'Total_Revenue': 'TOTAL_REVENUE'}, inplace=True)
```

In [30]:
```
print(df.columns.tolist())
```

```
['ORDERNUMBER', 'QUANTITYORDERED', 'PRICEEACH', 'ORDERLINENUMBER', 'SALES', 'ORDERDA
TE', 'STATUS', 'QTR_ID', 'MONTH_ID', 'YEAR_ID', 'PRODUCTLINE', 'MSRP', 'PRODUCTCOD
E', 'CUSTOMERNAME', 'PHONE', 'ADDRESSLINE1', 'ADDRESSLINE2', 'CITY', 'STATE', 'POSTA
LCODE', 'COUNTRY', 'TERRITORY', 'CONTACTLASTNAME', 'CONTACTFIRSTNAME', 'DEALSIZE',
'TOTAL_REVENUE']
```

In [ ]:
```

```